Why this challenge?

Germany has roughly 11,000 municipalities, and each runs its own dog-tax administration. For a citizen moving from Hannover to Leverkusen this means today: de-register in Hannover (form, letter, weeks of processing time), then register again in Leverkusen (form, letter, weeks of processing time). Two procedures, duplicate data entry, long redundancy.

The EU "once-only" principle (law in force since 2023) flips this around: data that a public authority already possesses does not have to be submitted again by the citizen – it is transferred in a controlled way between authorities.

You are building the SaaS platform that makes this possible: one platform, many municipalities, clean data separation - but with a controlled data flow exactly where the law permits it.

Task

Build a **SaaS-ready REST API** that provides three capabilities:

1. **Register a dog for tax purposes in a municipality (tenant).**
2. **Transfer an already-registered dog to another municipality when the owner moves.**
3. **Serve multiple municipalities simultaneously while keeping their data strictly isolated.**

Glossary

German term	English description
Halter / Steuerpflichtiger	Owner / taxpayer – the person who registers the dog and pays the tax
Veranlagung	Tax assessment – calculation of the tax amount for a calendar year
Mandant	Tenant – a municipality (e.g., "City of Berlin") that uses the SaaS platform

Listenhund	“Listed dog” – breeds classified as dangerous by law (e.g., Pitbull, Staffordshire Terrier); taxed at a higher rate and usually excluded from the normal tiered schedule
Ummeldung	Transfer of a dog-tax case from **Tenant A** to **Tenant B**, when the owner moves

What the system should do

New registration

- **Capture owner** – name, address
- **Capture dog** – name, chip number (mandatory), breed, type (`NORMAL` or `LISTENHUND`)
- **Automatically calculate tax** according to the statutes of the respective tenant
- **Assign the data to a tenant and store it in strict isolation**

Transfer (Ummeldung)

- Citizen moves from **Tenant A** to **Tenant B**
- Owner- and dog-data are handed over from **Tenant A** to **Tenant B**
- In **Tenant B** the tax is reassessed according to that tenant’s statutes
- **Tenant A** marks the case as “de-registered” (data stay retained, assessment ends)
- The hand-over is logged in a tamper-proof audit trail – who transferred which data to whom and when

Tax calculation

Choose **two real German dog-tax statutes** from the internet and implement their tiered rates. You need two statutes so that, after a transfer, the recalculation in the target tenant can be demonstrated. List the statutes you used in the final presentation.

Typical structure of a statute:

- Tiered basic tax depending on the number of dogs in a household (1st dog, 2nd dog, each additional dog)
- Higher tax rate for listed dogs

Simple `if/else` logic is sufficient.

Example API calls

Registration

Request

```
POST /anmeldung
X-Mandant-ID: hannover
Content-Type: application/json

{
  "halter": {
    "vorname": "Max",
    "nachname": "Mustermann",
    "strasse": "Hauptstraße",
    "hausnummer": 22,
    "plz": "30159",
    "ort": "Hannover"
  },
  "hund": {
    "name": "Bello",
    "chipnummer": "276099901234567",
    "rasse": "Labrador",
    "typ": "NORMAL"
  }
}
```

Response

```
{
  "status": 201,
  "personId": "...",
  "hundId": "...",
  "steuerbetrag": 96,
  "waehrung": "EUR",
  "veranlagungsjahr": 2026
}
```

Transfer (Ummeldung)

Request (sent to the receiving tenant)

```
POST /ummeldung
X-Mandant-ID: leverkusen
Content-Type: application/json

{
  "von_mandant": "hannover",
  "chipnummer": "276099901234567",
  "neue_adresse": {
    "strasse": "Am Büchelter Hof",
    "hausnummer": 5,
    "plz": "51379",
    "ort": "Leverkusen"
  }
}
```

Response

```
{
  "status": 200,
  "uebernommen_von": "hannover",
  "personId": "...",
  "hundId": "...",
  "neuer_steuerbetrag": 132,
  "waehrung": "EUR",
  "veranlagungsjahr": 2026,
  "protokoll_id": "..."
}
```